NEW YORK INSTITUTE OF TECHNOLOGY

College of Osteopathic Medicine

Learning Objectives:

1. Describe the steps that occur in primary and secondary hemostasis
2. Differentiate between the intrinsic, extrinsic, and common pathways of the coagulation cascade
3. Identify mechanisms of anticoagulation that prevent hypercoagulability states
4. Describe fibrinolysis as the final step in hemostasis
5. Apply the principle of hemostasis to clinical scenarios

Source: CPG

Question

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Which of the following has the correct receptor and ligand pair?

a) GpIb receptor and endothelin
b) P2Y12 receptor and Von Willebrand Factor
c) P2Y12 receptor and calcium
d) GpIIbIIIa receptor and fibrin
e) GpIIbIIIa receptor and fibrinogen

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A young girl falls and scraps her knee. After the platelet plug is formed, what pathway will be activated next to stabilize the platelet plug?

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b) Intrinsic pathway
c) Extrinsic pathway
d) Common pathway
e) Fibrinolysis

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What is the main role of antithrombin?

a) Positive feedback on the coagulation cascade
b) Bind and activate thrombin
c) Bind and degrade thrombin
d) Bind thrombin, protein C and protein S to inactivate clotting factors
e) Bind fibrinogen and convert it to fibrin

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Question

A 50-year-old man comes into the ED complaining of black stools for the past week. When you ask about his diet, he tells you that he is a very picky eater and eats a lot cereal, grains and potatoes. He does not eat much of meat, fruits or vegetables. You think that he may have a vitamin deficiency that is making him bleed. Which of the following has been depleted due to his poor diet?

a) Platelets
b) Von Willebrand Factor
c) Calcium
d) Factors V and X
e) Factors II and VII

Question

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d) Factors V and X
e) Factors II and VII

Vit K dependent= factors II, VII, IX, X, protein C and S

Summary

1. Injury
2. Primary hemostasis
    1. Vasoconstriction
    2. Platelet plug
        1. Adhesion
        2. Activation
        3. Aggregation
3. Secondary hemostasis
    1. Intrinsic vs extrinsic
    2. Common pathway
4. Fibrinolysis

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